

ON THE VIETORIS POWER OF A CONVERGENT SEQUENCE

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ABSTRACT. We prove that $K(\omega + 1, \text{ord})$ is second-countable, hence Lindelöf, but is not σ -compact. This answers the $\omega + 1$ case of Question 1 on page 11 of Caruvana and Holshouser, arXiv:2507.17936v3. A countable family of range and coordinate restrictions proves the positive result. The negative result follows from finite occurrence bounds on compact families of surjections and a delayed diagonal enumeration. The complete argument has been formalized and checked in Lean 4, including its realization in the order topology.

Key words and phrases. Vietoris power; ordered compact sets; convergent sequence; second countability; Lindelöf space; σ -compactness; Lean.

1. THE PROBLEM AND THE PRECISE SPACE

Caruvana and Holshouser [1, Question 1, p. 11] ask whether the ordered compact-set space $K(\alpha, \text{ord})$ is σ -compact or Lindelöf for countable ordinals $\omega < \alpha < \omega_1$, and explicitly ask about $\alpha = \omega + 1$. All page and definition numbers below refer to arXiv version 3, not to another edition.

Theorem 1. With its Vietoris-power topology, $K(\omega + 1, \text{ord})$ is second-countable and Lindelöf, and is not σ -compact.

Write $X = \omega + 1$, with the order topology; the point ω is the limit of the isolated natural numbers. In [1, Definition 1.8 and Proposition 1.9, p. 4], basic sets in the Vietoris power are

$$[U; F, V] = \{f \in X^\kappa : \text{img}(f) \subseteq U, f(i) \in V_i \ (i \in F)\},$$

where U and the sets V_i are open in X and $F \subseteq \kappa$ is finite. Definition 1.14 [1, p. 5] restricts this topology to the functions whose images are compact, taking κ to be the supremum of the cardinalities of compact subsets, plus ω in cardinal arithmetic.

Here X itself is countably infinite and compact, so that cardinal is ω . Consequently the exact space considered in this paper is

$$K = \{f : \omega \rightarrow X : \text{img}(f) \text{ is compact}\},$$

with the subspace topology generated by the displayed basic sets. The topology on K includes a restriction on the entire image as well as finitely many coordinate restrictions. We address only the case $\omega + 1$ of the source question.

2. A COUNTABLE GENERATING FAMILY

Lemma 2. A compact subset of X avoiding ω is finite. Every open subset of X containing ω has finite complement.

Proof. A subset avoiding ω is covered by its open singleton subsets. If it is compact, this cover has a finite subcover. If an open set contains ω , its complement is closed in the compact space X and avoids ω , so the first assertion applies. Conversely, a finite subset of ω is open; every cofinite subset of X is open because X is a Hausdorff space.

Let A be the countable family consisting of all finite open subsets of X and all cofinite open subsets of X . For $U \in A$ and $i \in \omega$, set

$$T(U) = \{f \in K : \text{img}(f) \subseteq U\},$$

$$E(i, U) = \{f \in K : f(i) \in U\}.$$

These sets form a countable subbase for the topology of K . To verify that they suffice, consider first an arbitrary open range bound U . If $\omega \in U$, Lemma 2 puts U in A . Otherwise, any $f \in T(U)$ has finite image, and its image R is an open member of A with $f \in T(R) \subseteq T(U)$. Thus every open range restriction is a union of restrictions from our family.

For an arbitrary open coordinate bound U , the same argument applies when $\omega \in U$. If ω is absent and $f(i) \in U$, the singleton $\{f(i)\}$ is open and belongs to A ; the corresponding coordinate condition refines $E(i, U)$ at f . Finally, each original basic set is the intersection of one range condition and finitely many coordinate conditions. Finite intersections of a countable subbase form a countable base. This proves second countability and hence Lindelöfness.

3. CLOSED SURJECTIONS AND COMPACT OCCURRENCE BOUNDS

For the negative assertion, define

$$S = \{f \in K : f(0) = \omega, \text{img}(f) = X\}.$$

Lemma 3. The set S is closed in K .

Proof. Coordinate evaluations are continuous, so the condition $f(0) = \omega$ is closed. For each $x \in X$, the complement of the condition $x \in \text{img}(f)$ is the open tube $T(X \setminus \{x\})$. Surjectivity is the intersection, over all $x \in X$, of these closed conditions. Their intersection with the zeroth-coordinate condition is S .

Lemma 4. If $C \subseteq S$ is compact and $n \in \omega$, there is a natural number b such that each $f \in C$ takes the value n at some coordinate $k \leq b$.

Proof. The open sets $\{f \in K : f(k) = n\}$, indexed by $k \in \omega$, cover C , since its members are surjective and n is isolated. A finite subcover exists. Let b be the largest selected coordinate, taking $b = 0$ when the selected family is empty. This gives the required bound.

4. A DELAYED DIAGONAL ENUMERATION

Proof of the negative assertion in Theorem 1. Suppose K is σ -compact. Intersecting a countable compact cover with the closed set S gives compact sets $C_n \subseteq S$ whose union is S . By Lemma 4, for each n choose $b(n)$ such that every member of C_n takes value n at some coordinate at most $b(n)$.

Choose distinct positive coordinates $j(n)$ with $j(n) > b(n)$. For example, choose them recursively, each larger than the corresponding bound and all earlier chosen coordinates. Define $g : \omega \rightarrow X$ by

$$g(j(n)) = n \ (n \in \omega), \ g(k) = \omega \ (\text{all other } k).$$

The map g has $g(0) = \omega$ and is surjective. Its image is the compact space X , so $g \in S$. But the only coordinate at which g takes value n is $j(n) > b(n)$. Hence g belongs to no C_n , contradicting their covering property. This proves that K is not σ -compact and completes Theorem 1.

In the Lean implementation, the recursive choice of coordinates is replaced by $j(n) = p(n, b(n)+1)$, where p is the standard natural-number pairing bijection used by `mathlib`. Its inverse has second coordinate at most the original index. Therefore the chosen positions are distinct, positive and exceed the respective bounds. This is the same diagonal argument.

5. FORMAL VERIFICATION AND SCOPE

The accompanying Lean project defines the topology from the source basic sets and defines the carrier by actual compactness of the image. It proves second countability, Lindelöfness and failure of σ -compactness without hypotheses assuming these conclusions. The initial ground space is the one-point compactification of discrete natural numbers.

An identity homeomorphism identifies that space with the extended natural numbers in their order topology. Equality of the ground topologies then transfers the entire compact-range construction. This checks the concrete order-topological realization of $\omega + 1$; it does not use a subtype of `mathlib` ordinals.

Checked declaration: `VietorisOmegaOne.order_model_theorem`.

The project was checked with Lean 4.19.0 and `mathlib` v4.19.0. The transitive axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. No admitted proof, custom mathematical axiom or native-computation trust axiom occurs in the final theorem. The companion verification record includes the audit output, pinned dependencies and file hashes. Identifying the source question with these formal definitions remains an explicit mathematical reading of [1].

AI disclosure. ChatGPT Astra performed almost all of the work, including the mathematical development, Lean formalization and preparation of the manuscript. The author reports that his role was to check and edit the paper. This disclosure does not assert independent peer review.

REFERENCE

- [1] Christopher Caruana and Jared Holshouser, An adaptation of the Vietoris topology for ordered compact sets, arXiv:2507.17936v3 (10 November 2025). Definitions 1.8 and 1.14, pp. 4-5; Question 1, p. 11.
<https://arxiv.org/abs/2507.17936v3>